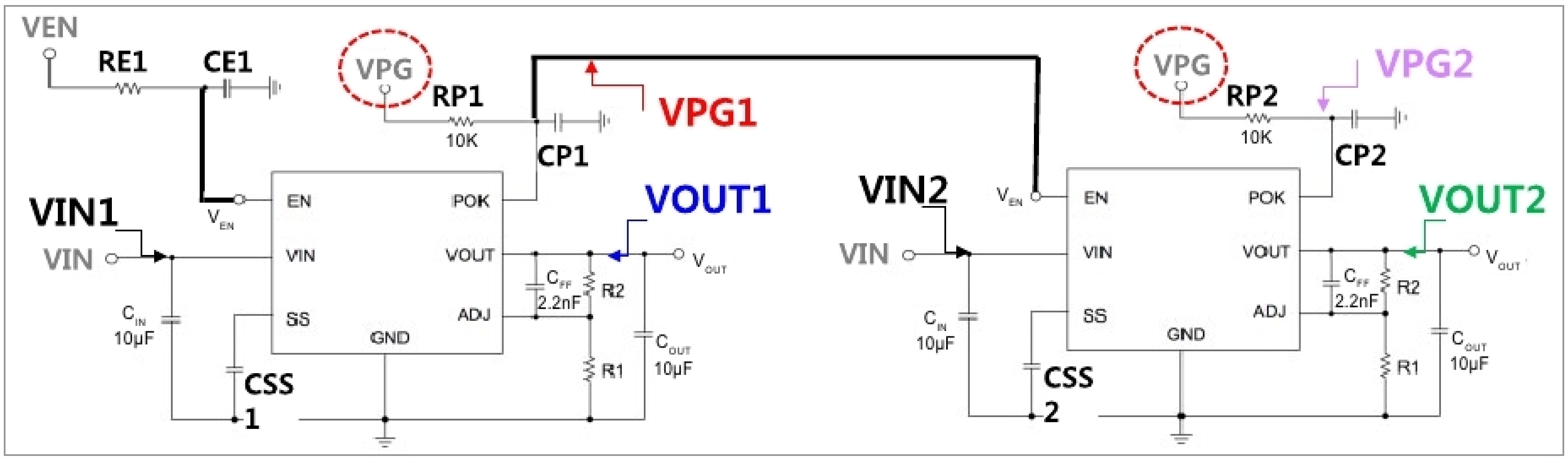

Fig. 8: Application example of PowerGood signal for power sequencing

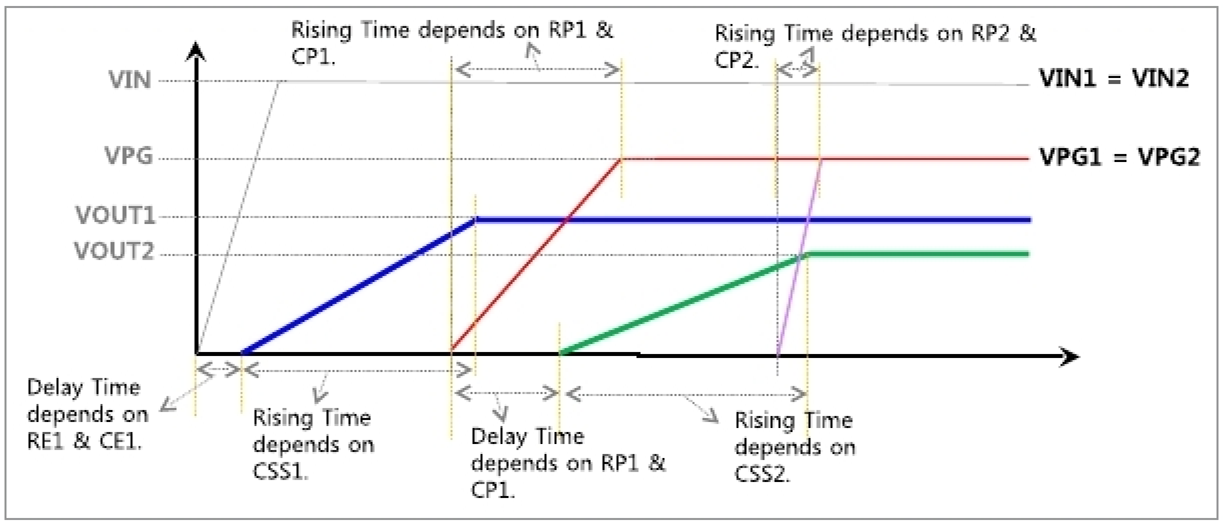

Fig. 9: Power sequence timing diagram of Fig. 8

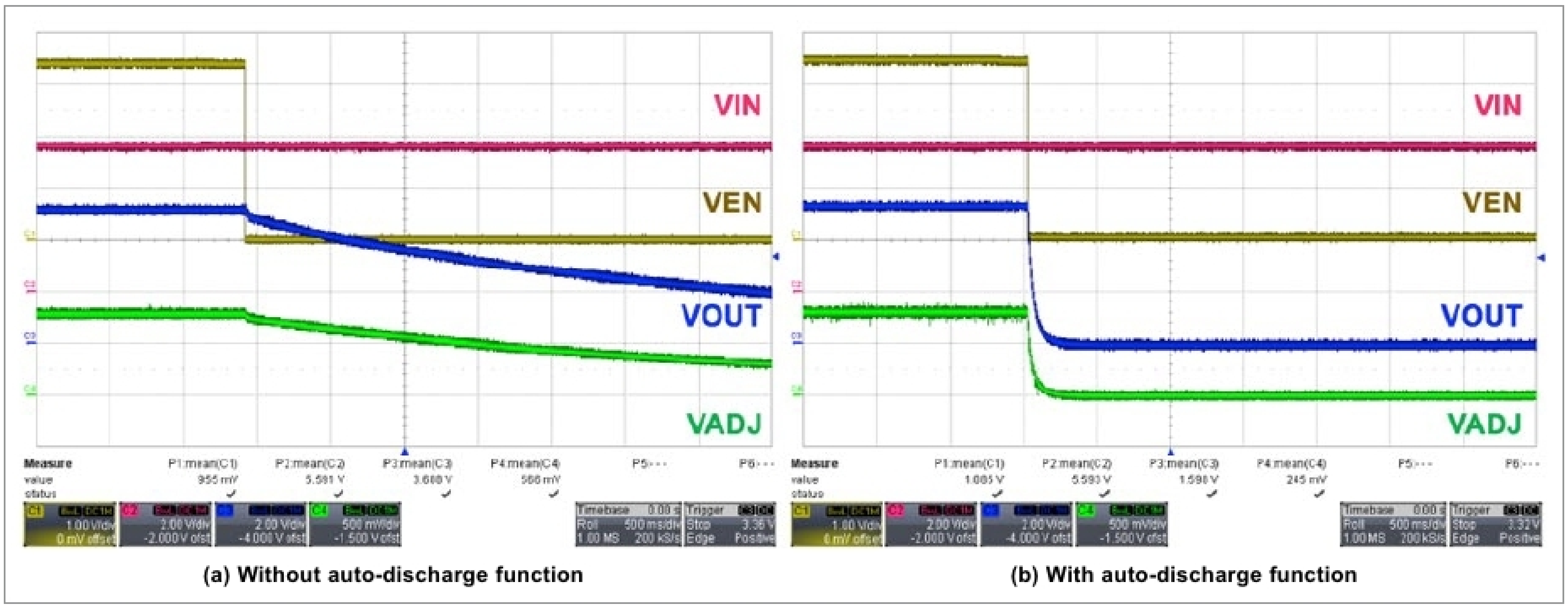

Fig. 10: Power off discharge characteristics

sion performance of PMOS LDO. The typical block diagram is shown in Fig. 3(b), and on-resistance performance is also proportional to V_{IN}, resulting in high V_{GS}, but V_{GS} can be changed depending on V_{OUT}.

So, NMOS LDO has independent voltage supply V_{BIAS} for the internal circuit to remove V_{IN} restriction in dropout voltage performance when output voltage is low. As shown in Fig. 4(b), it maintains low dropout voltage performance even though output voltage is low. It means that V_{IN} can be $[V_{OUT} + V_{DROPOUT}]$ without any minimum input voltage restriction like PMOS LDO.

The disadvantage of NMOS LDO is the existence of V_{BIAS}, the power supply for internal control circuit. To make NMOS pass element have appropriate on-resistance, V_{BIAS} is higher (generally $3V_T$) than V_{OUT}. Now, NMOS LDO has maximum V_{BIAS} restriction because of maximum operating rating of V_{BIAS} pin.

When V_{OUT} is getting higher, V_{BIAS} should also be getting higher to make NMOS pass element have suitable on-resistance. But, unfortunately, maximum allowable voltage is restricted by fabrication process. So, due to maximum V_{BIAS} voltage restriction, maximum output voltage performance is restricted by V_{BIAS} rating. As a result, we can conclude that PMOS LDO is suitable for relatively higher output voltage applications, whereas NMOS LDO is suitable for low voltage to low voltage conversions, as shown in Fig. 4. This means that stability and reliability of the power circuit depend not only on component performance but also on proper selection of components.

Additional functions: soft start, power-good and auto-discharge

General protection functions such as output over-current protection